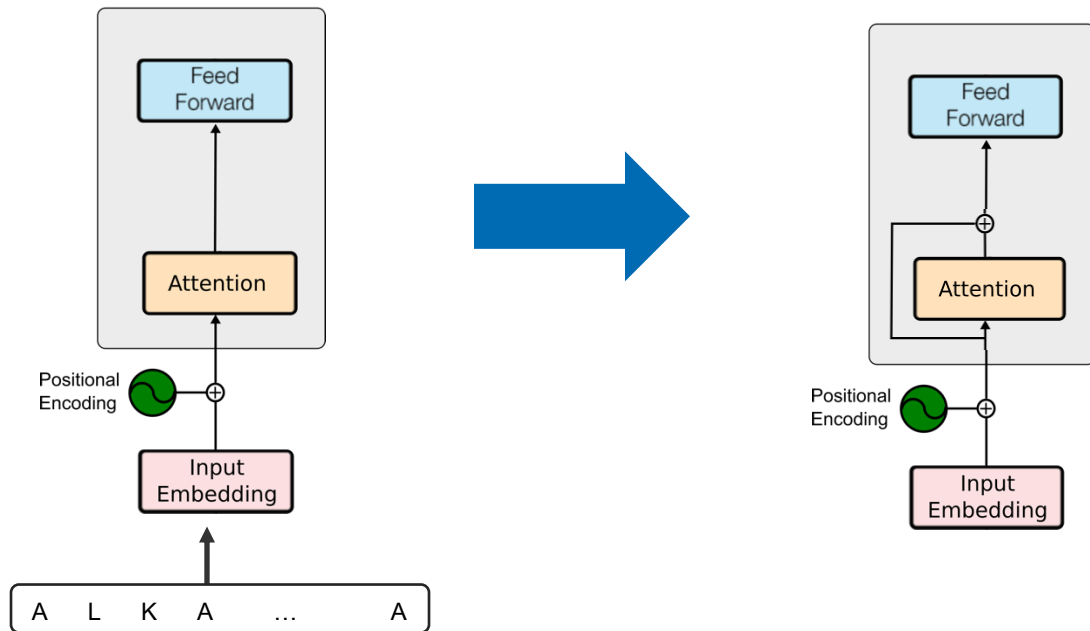
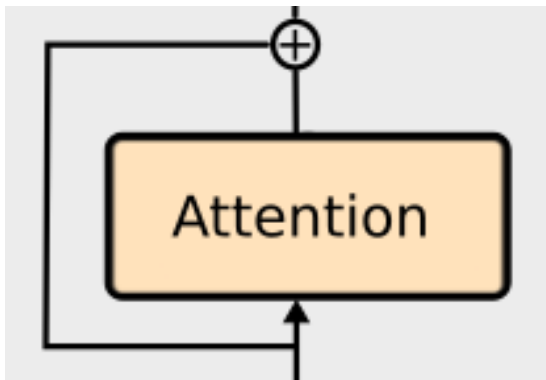


Transformer Network Encoders

A more detailed view



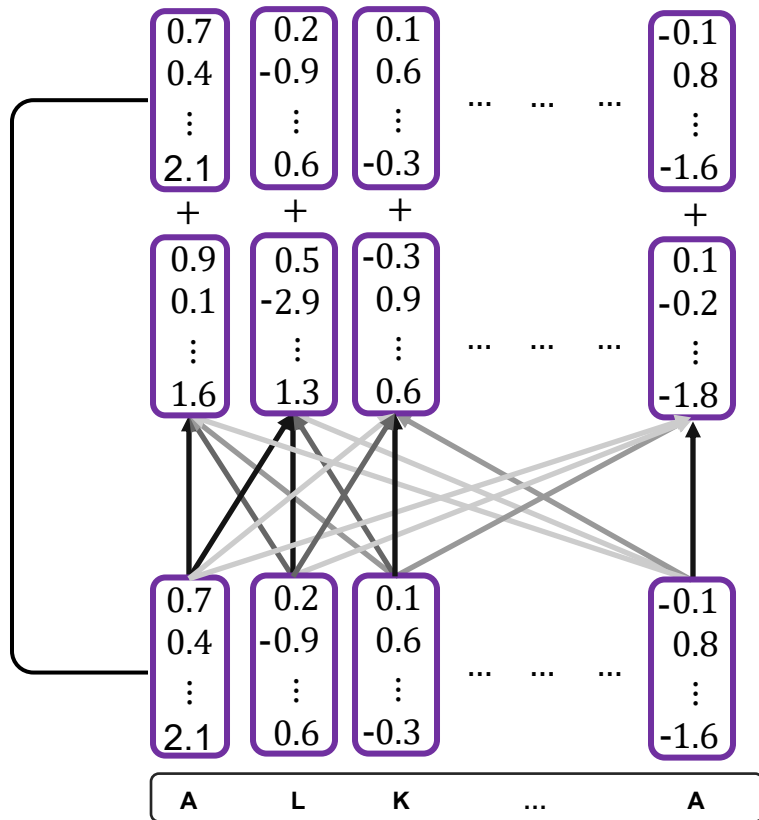
Residual connection

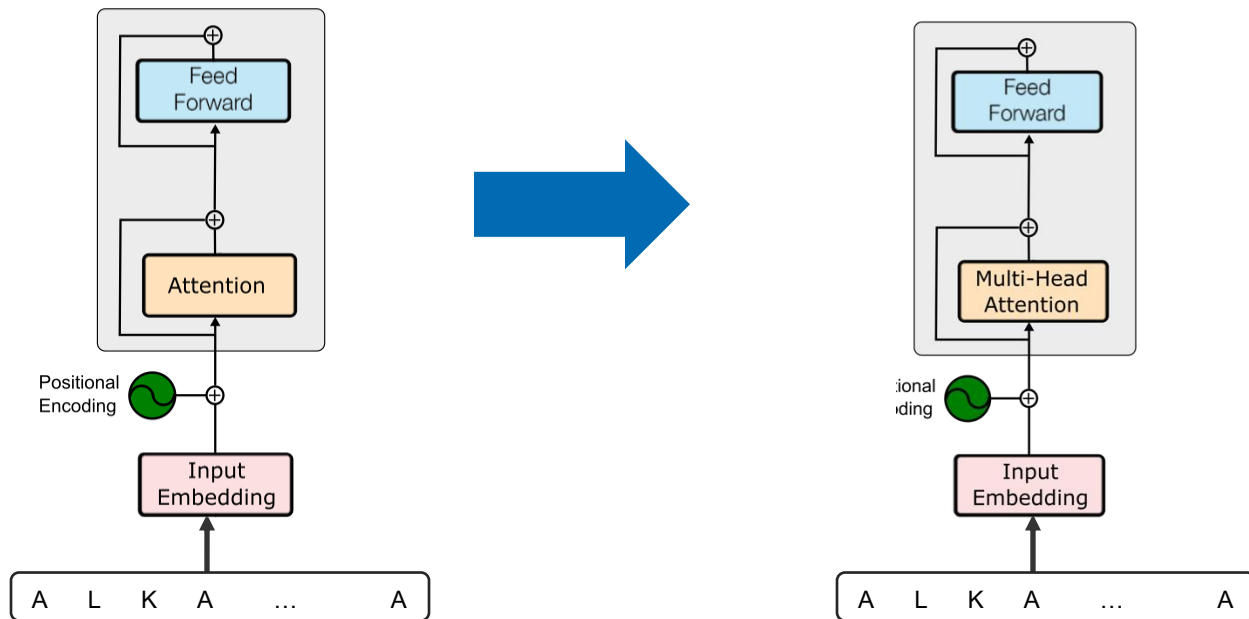


$$\vec{e}_i = \text{Attention}(\vec{e}_i, K, V)$$

Adding residual
connection

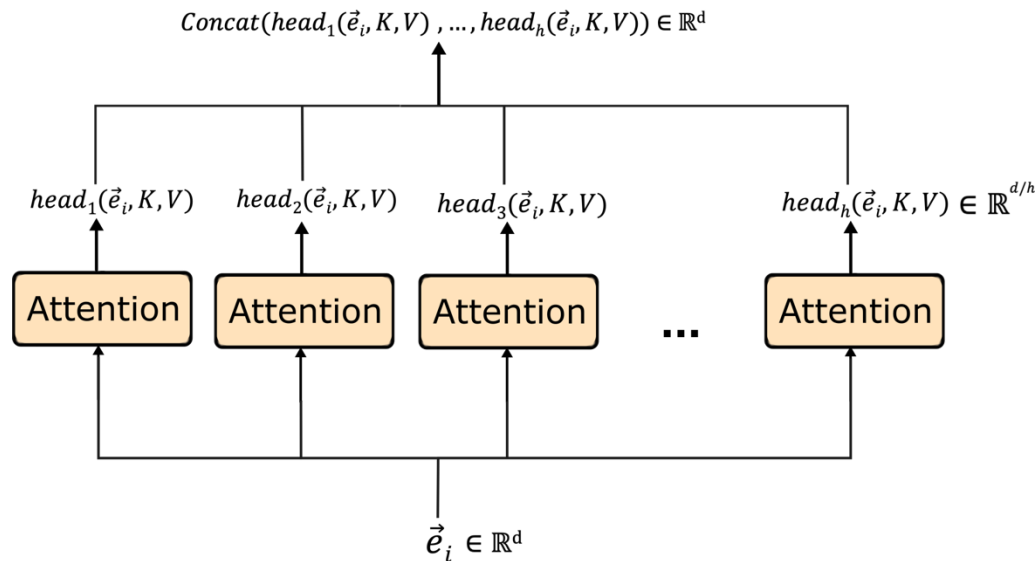
$$\vec{e}_i = \text{Attention}(\vec{e}_i, K, V) + \vec{e}_i$$





Multi-Head Attention

- Instead of applying attention function once, we apply it h times in parallel, e.g., $h = 8$.

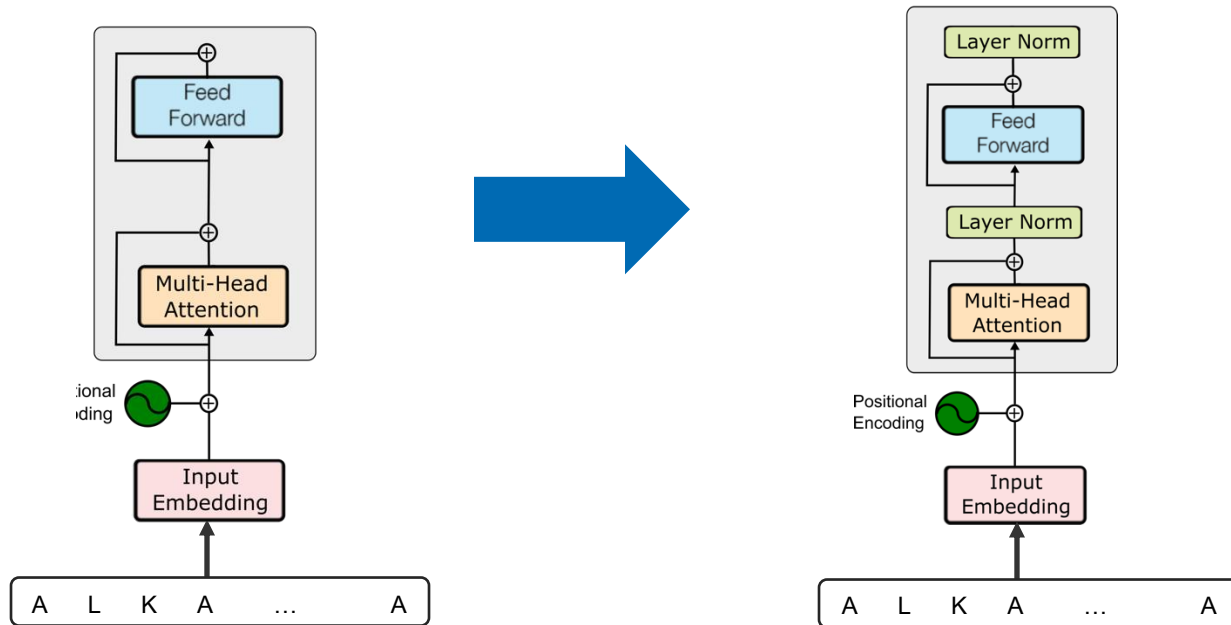


- Each vector produced by one of the h attention heads should have dimension $m := d/h$

- $W_E^j, W_K^j, W_V^j \in \mathbb{R}^{m \times d}$
- $W_O \in \mathbb{R}^{d \times d}$

$$head_j(\vec{e}_i, K, V) = Attention(W_E^j \cdot \vec{e}_i, W_K^j \cdot K, W_V^j \cdot V), \quad j = 1, \dots, h; \quad \vec{e}_i \in \mathbb{R}^d$$

$$MultiHead(\vec{e}_i, K, V) = W_O \cdot Concat(head_1(\vec{e}_i, K, V), \dots, head_h(\vec{e}_i, K, V)) \in \mathbb{R}^d$$



Layer Normalization

- Example: $d = 3$, number of tokens in input sequence $N = 2$, and batch size 1

- Output of Multi-Head attention

$$\vec{e}_1 = \begin{pmatrix} 0.5 \\ -3 \\ 8.5 \end{pmatrix} \quad \vec{e}_2 = \begin{pmatrix} -2 \\ 4 \\ 1 \end{pmatrix}$$

- Calculating mean and standard deviation for each sample:

$$\mu_1 = \frac{1}{3} (0.5 - 3 + 8.5) = 2, \quad \mu_2 = \frac{1}{3} (-2 + 4 + 1) = 1$$

$$\sigma_1 = \sqrt{\frac{1}{3} [(0.5 - \mu_1)^2 + (-3 - \mu_1)^2 + (8.5 - \mu_1)^2]} = 2$$

$$\sigma_2 = \sqrt{\frac{1}{3} [(-2 - \mu_2)^2 + (4 - \mu_2)^2 + (1 - \mu_2)^2]} = 2$$

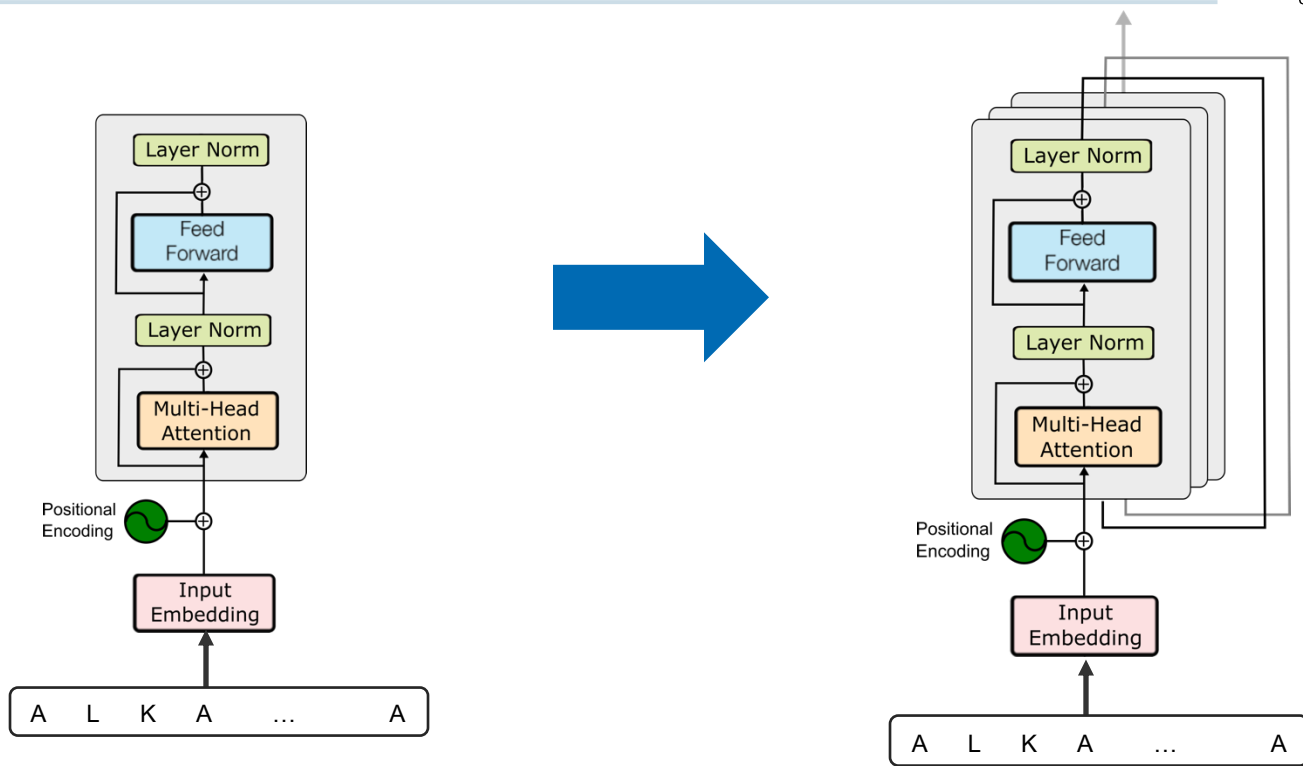
- Normalizing all vectors

$$\frac{\vec{e}_1 - \mu_1}{\sigma_1} = \begin{pmatrix} -0.75 \\ -2.5 \\ 3.25 \end{pmatrix} \quad \frac{\vec{e}_2 - \mu_2}{\sigma_2} = \begin{pmatrix} -1.5 \\ 1.5 \\ 0 \end{pmatrix}$$

- Adding learnable rescaling parameters $\gamma, \beta \in \mathbb{R}^3$

$$\gamma \odot \left(\frac{\vec{e}_1 - \mu_1}{\sigma_1} \right) + \beta \quad \gamma \odot \left(\frac{\vec{e}_2 - \mu_2}{\sigma_2} \right) + \beta$$

Encoder



Parallelization of Attention Calculations

$$\text{Attention}(\vec{e}_i, K, V) = \text{softmax}\left(\frac{\vec{e}_i^T \cdot K}{\sqrt{d}}\right) \cdot V^T \in \mathbb{R}^d \quad K = (\vec{e}_1, \vec{e}_2, \dots, \vec{e}_7),$$

$$V^T = \begin{pmatrix} \vec{e}_1^T \\ \vec{e}_2^T \\ \vdots \\ \vec{e}_7^T \end{pmatrix}$$

Parallelize

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q^T \cdot K}{\sqrt{d}}\right) \cdot V^T \in \mathbb{R}^{N \times d}$$

$$Q^T = \begin{pmatrix} \vec{e}_1^T \\ \vec{e}_2^T \\ \vdots \\ \vec{e}_7^T \end{pmatrix}$$

$$\text{Attention}(W_E \cdot Q, W_K \cdot K, W_V \cdot V)$$